When does an LLM autoresearch loop find real signal?

A controlled study on twelve biomedical prediction tasks

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Abstract

Autoresearch loops, in which a large language model (LLM) repeatedly rewrites a single solution file and a change is kept only if a fixed development score improves, are an increasingly common way to build predictive pipelines. We ask when such a loop finds signal that holds up on held-out and external data. We ran a loop driven by Claude Opus 5.5 on 12 biomedical prediction tasks (scalp and neonatal EEG, resting-state fMRI, 16S microbiome, proteomics and drug response; 3 splits per task, 20 rounds per loop). Each champion was scored once on a held-out 25% of subjects and on eight external datasets, and compared with three baselines given the same 20 development evaluations: one LLM proposal, 20 independent LLM proposals without feedback, and Optuna search over a pipeline written with knowledge of the loop's findings. Development scores improved in all 36 runs; held-out scores in 25. Averaged over 11 AUROC tasks the loop gained +0.050 AUROC over the seed (95% CI +0.006 to +0.094), compared with +0.004 for one proposal, +0.014 for 20 proposals without feedback and +0.033 for Optuna. The pooled gain was not robust to removing any one of six tasks, and the loop's advantage over Optuna (+0.017, $p = 0.28$) was significant only for abnormal-EEG and neonatal-seizure detection, the two tasks whose gains also replicated across independent repeats of the loop. When site and demographics were available, the loop built its models on them, and its gains on psychiatric EEG did not transfer to external cohorts. Feedback is what makes the LLM useful here, but the loop does not replace shortcut-free scoring and external validation.

1 Introduction

LLM agents can now write, run and revise machine-learning code with little supervision [10, 22, 23, 26]. The simplest version is the *autoresearch loop* [25, 32]: one editable file, one fixed metric, and a model that proposes a change each round, kept only if the metric improves. On development data the loop always looks like it works, because a keep-if-better rule over a fixed cross-validation (CV) score cannot make that score go down.

Biomedical prediction is a hard test of whether such gains are real. Datasets are small, cases and controls are often recruited and recorded differently, and site, scanner, batch and demographics can predict the label better than the biology [17, 37, 38]. Leakage and shortcuts are already common in human-written studies [24, 31], and selecting the best of many candidates on one CV score is itself a source of optimistic bias [9]. An optimizer that tries many ideas against one score has every opportunity to find all of these.

We therefore ask not whether the loop improves the development score, but *when* the improvement holds up: on held-out subjects, on external cohorts, against simpler uses of the same budget, and under checks for confounding and leakage. Across 12 tasks in six domains under one fixed protocol, we contribute (i) held-out results for 36 loop runs, each scored once, plus independent repeats on 9 tasks; (ii) three same-budget baselines that separate the effect of feedback from that of

the LLM and of ordinary search; (iii) multiplicity-corrected per-task statistics that include split and run variation, and pooled estimates with sensitivity analyses; (iv) evidence that the loop exploits site and demographic shortcuts when they are available, and an age/sex confound analysis of the shortcut-free results; and (v) transfer to eight external datasets, a transduction audit and a leakage audit.

2 Related work

Benchmarks such as MLAgentBench [22] and MLE-bench [10] evaluate agents that write and run training code; AIDE [23] searches code with tree-structured refinement; the AI Scientist [26] automates the wider research cycle; and FunSearch [33] and AlphaEvolve [27] pair LLM proposals with an automatic evaluator. Most of this work reports scores on the benchmark the agent optimized. The autoresearch loop [25] is a deliberately minimal member of this family, and our protocol reuses that of radnyees [32]. As a non-LLM reference we use tree-structured Parzen estimators [7] in Optuna [2]. Our focus is a controlled evaluation in a domain where selection bias [9], leakage [24] and site or demographic shortcuts [8, 38] are well documented.

3 Methods

Loop. A fixed harness loads the data and defines labels, splits and metric; it is never shown to or edited by the LLM. A solution is one Python file implementing `featurize(items, ctx)`, which never receives labels, and `fit_predict(F_train, y_train, F_test, ctx)`, which receives training labels only. Each round the LLM (Claude Opus 5.5, reasoning effort *high*) receives a general brief, a project brief, the current best solution and the full history of previous rounds (rationale, declared feature/model families, score, kept or not, last failure), and returns a complete new solution. Candidates run on CPU containers with a 1,500 s limit (timeouts and crashes count as failed rounds). A loop runs 20 rounds with patience 10 and keeps a candidate iff its development score improves by more than 10^{-4}.

Tasks. Table 1 lists the 12 tasks: abnormal EEG (TUAB [28]); depression and ADHD (TDBRAIN [36]; the two tasks share 90 controls); schizophrenia (ASEEG [3]); per-second neonatal seizure detection (Helsinki [34]); autism and ADHD from fMRI connectivity (ABIDE I [15], ADHD-200; [29]); colorectal cancer (CRC) and IBD from 16S stool profiles (MicrobiomeHD [5, 16]); Alzheimer's disease (AD) from plasma [30] and CSF [4] proteomics; and drug response from CCLE expression to PRISM AUC [11, 13]. Seeds were written before any loop ran and are deliberately standard: log band power with L2 logistic regression (scalp EEG); line length, RMS and band power with gradient boosting and smoothing (neonatal EEG); Fisher-z CC200 connectivity with logistic regression (fMRI); genus abundance with a random forest (16S); $\log_2$, median centring and logistic regression (proteomics); and ridge regression on the most variable genes (drug response).

Shortcut-free protocol (v2). Our first protocol (v1) passed age, sex, head motion and site to the solution and scored multi-site tasks with pooled AUROC; the loop used them (Section 4.4). All other results use v2: no demographics or motion; site, study and batch only for harmonization; multi-site tasks scored *within site* (size-weighted mean of per-site AUROCs); and TDBRAIN tasks restricted to the age range shared by patients and controls. v2 was designed after we saw v1 results, and the v2 loops were rerun on the same split seeds; this and other post-hoc decisions are listed in Appendix B.

Table 1: Tasks. Eval n = held-out units in split 1 (recordings for TUAB; stool samples for IBD, grouped by person or twin pair); other splits differ by at most a few. †Size-weighted mean of per-site (or per-study) AUROCs.

Task	Data	Label	Metric	Eval n (cases)	External sets
Abnormal EEG	TUAB v3.0.1, train split	abnormal vs. normal	AUROC	688 rec. / 519 pat. (346)	TUAB official eval
MDD	TDBRAIN, age-matched	MDD vs. healthy	AUROC	70 (38)	Mumtaz 2016
ADHD	TDBRAIN, adults	ADHD vs. healthy	AUROC	43 (13)	–
Schizophrenia	ASEEG	SZ vs. healthy	AUROC	25 (13)	MSU, RepOD
Neonatal seizure	Helsinki	seizure second (2/3 experts)	concat. AUROC	20 neonates	–
Autism	ABIDE I, 20 sites	ASD vs. TD	AUROC†	262 (129)	–
ADHD	ADHD-200, 8 sites	ADHD vs. TD	AUROC†	217 (82)	–
CRC	MicrobiomeHD, Baxter	CRC vs. healthy	AUROC	73 (30)	Zeller, Zackular, Xiang, Zhao
IBD	MicrobiomeHD, 4 studies, stool	IBD vs. control	AUROC†	113 samples / 103 groups (91)	–
AD, plasma	PXD065835	AD vs. control	AUROC	22 (13)	–
AD, CSF	Bader 2020, 4 cohorts	clinical AD vs. non-AD	AUROC†	34 (12)	–
Drug response	CCLE → PRISM, 200 drugs	dose–response AUC	per-drug Pearson r	120 cell lines	–

Evaluation. For each split seed (1, 2, 3), 25% of subjects (grouped by subject; stratified by label and site) form a held-out *Eval* set and the rest a *dev pool*. The loop optimizes *Dev*, a 5-fold grouped CV score inside the dev pool. After the loop, seed and champion are refit on the whole dev pool and scored *once* on Eval and on external datasets. The three Eval sets overlap, so splits are not independent replicates. On 9 tasks, split 1 was rerun twice more from scratch (only the LLM's sampling differs) to measure run-to-run variation.

Same-budget baselines. All arms start from the same seed, use the same harness, time limit and 20 Dev evaluations, keep a candidate only if it beats the seed on Dev, and are scored once on the same Eval sets. *One-shot*: one LLM proposal. *Best-of-20*: 20 independent LLM proposals that see the briefs and the seed but no history or scores, isolating the value of feedback. *Optuna*: 20 trials of multivariate TPE over a hand-written parameterized pipeline per project (which feature blocks to include, which classifier, its regularization). The search spaces were written *after* the main study and include feature families the loop had found, so this baseline is informed by the loop's results: stronger than an independent search, and conservative from the loop's point of view. Arms are equalized in Dev evaluations, not in compute or LLM tokens.

Statistics. Per task, the gain is the mean over three splits. We report three two-sided p-values (2,000 bootstrap draws, $(k+1)/(n+1)$ convention [12], Holm-adjusted across 12 tasks [21]): a *subject* bootstrap that resamples subjects and treats the fitted models as fixed; a *hierarchical* bootstrap that resamples splits, then loop runs where repeats exist, then subjects; and a t-test across the three split gains. Per-task claims use the hierarchical p. With only three splits its split level is coarse: when all three split gains have the same sign, resampling splits cannot produce a mean of the opposite sign, so the hierarchical p then largely reflects sign consistency plus subject noise. We therefore also report sign agreement across independent repeats. Pooled over the 11 AUROC tasks we report the equal-weight mean of task gains with a t-interval (primary) and DerSimonian–Laird random effects [14] (secondary), with leave-one-task-out and confound-based sensitivity analyses. Details of the shortcut-only models, the confound analysis and the transduction and leakage audits are in Appendix A.

Table 2: Held-out Eval per arm with the same budget of 20 Dev evaluations (bold: best non-seed arm), and the loop's gain over the seed and over Optuna with Holm-adjusted hierarchical bootstrap p-values (bold gain: $p_{\text{Holm}} < 0.05$; ^: significant only conditionally on the fitted models). splits ↑: splits where the loop beat the seed on Eval. †Within-site AUROC. ‡Mean per-drug Pearson r. §Gain carried by split 1, whose seed was below chance (split gains +0.264, +0.001, +0.015).

	Held-out Eval (mean of 3 splits)					Loop − seed		Loop − Optuna		
Task	Seed	One-shot	Best-of-20	Optuna	Loop	gain	p_{Holm}	gain	p_{Holm}	splits ↑
Abnormal EEG (TUAB)	0.847	0.866	0.888	0.889	**0.918**	**+0.071**	0.012	**+0.029**	0.012	3/3
MDD (TDBRAIN)	0.850	0.864	0.884	0.946	**0.954**	**+0.104**	0.012	+0.008	1.000	3/3
ADHD (TDBRAIN)	0.785	0.785	0.785	0.918	**0.922**	**+0.137**	0.036	+0.004	1.000	3/3
Schizophrenia (ASEEG)	0.677	0.677	0.705	0.678	**0.799**	+0.122	0.480	+0.121	0.440	3/3
Neonatal seizure (Helsinki)	0.827	0.827	0.827	0.824	**0.899**	**+0.072**	0.012	**+0.075**	0.012	3/3
Autism (ABIDE I)†	0.781	0.783	**0.790**	0.781	0.778	−0.002	1.000	−0.003	1.000	1/3
ADHD (ADHD-200)†§	0.572	0.570	0.627	0.650	**0.665**	+0.093^	0.935	+0.015	1.000	3/3
CRC (Baxter)	0.795	**0.789**	0.787	0.759	0.740	−0.055^	0.480	−0.020	1.000	0/3
IBD (4 studies)†	0.721	**0.755**	0.750	0.688	0.735	+0.014	1.000	+0.047	1.000	1/3
AD, plasma	0.912	0.895	0.897	**0.934**	0.866	−0.046	0.930	−0.068	1.000	0/3
AD, CSF†	0.913	0.913	0.891	**0.971**	0.954	+0.041	0.935	−0.017	1.000	2/3
Drug response (PRISM)‡	0.149	0.149	0.149	0.148	**0.160**	+0.012	0.935	+0.013	1.000	3/3

4 Results

4.1 Development gains are guaranteed; held-out gains are not

Across 36 runs the loop ran 710 rounds and kept 281 candidates (7 failed rounds). Dev improved in 36/36 runs, as the keep-if-better rule guarantees; Eval improved in 25/36, with per-split $p < 0.05$ in 15/36 (Table 4). The mean Dev gain (0.075) exceeded the mean Eval gain (0.047) and the two were only moderately correlated ($r = 0.37$; Figure 6). CRC, plasma AD, ABIDE autism and IBD lost or stayed flat on Eval in at least two of three splits despite consistent Dev gains (Table 2).

4.2 Feedback, not the LLM alone, produces the gains

Table 2 and Figure 1 compare the loop with the baselines. One LLM proposal was kept in 17/36 splits and changed Eval by +0.004 on average. Twenty proposals without feedback did better (+0.014) but were kept in only 25/36 splits: on neonatal seizure detection and ADHD (TDBRAIN), none of 120 independent proposals beat the seed on Dev, while the loop improved every split of both tasks. The loop's own first two rounds on neonatal seizures also failed in every split, and its first kept change came in round 3. All 720 best-of-20 proposals ran, so low adoption reflects proposal quality, not a broken baseline.

The loop-informed Optuna search gained +0.033 on average ($p = 0.073$). It matched the loop on both TDBRAIN tasks (MDD 0.946 vs. 0.954; ADHD 0.918 vs. 0.922) and on ADHD-200 (0.650 vs. 0.665), and beat it on plasma and CSF AD; on IBD a single LLM proposal did best. The loop's advantage over Optuna was Holm-significant only on abnormal EEG (+0.029) and neonatal seizure detection (+0.075).

4.3 Per-task and pooled statistics

With the hierarchical bootstrap, the loop's gain over the seed is Holm-significant on four tasks: abnormal EEG (+0.071), MDD (+0.104) and ADHD (+0.137) on TDBRAIN, and neonatal seizure

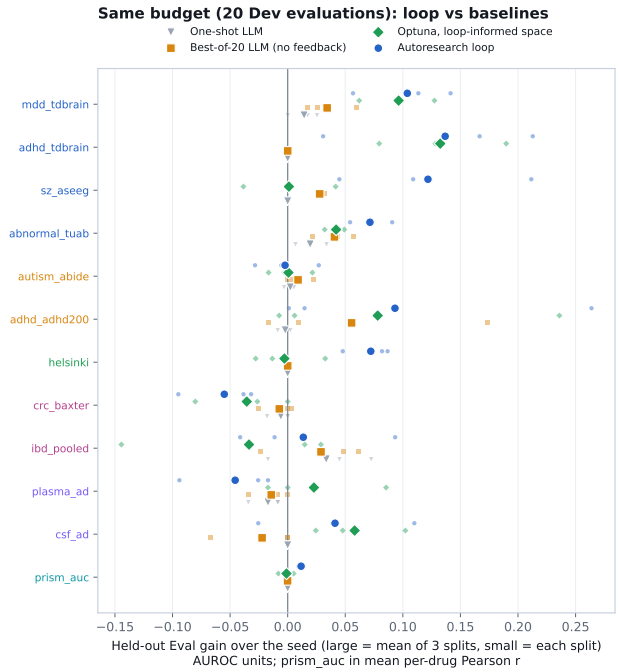


Figure 1: Held-out Eval gain over the seed per arm, same budget. Large markers: mean of 3 splits; small markers: individual splits.

detection (+0.072) (Table 2). The CRC loss (−0.055) and the ADHD-200 gain (+0.093) are significant only conditionally on the fitted models; the ADHD-200 gain rests on one split. The caveat in Section 3 applies to the four significant tasks: their three split gains all have the same sign, and the t-test across splits, which does not have that problem but has little power, gives $p = 0.021$ (abnormal EEG), 0.028 (neonatal seizure), 0.053 (MDD) and 0.13 (ADHD on TDBRAIN). Independent repeats are the stronger evidence. On abnormal EEG, neonatal seizure detection and MDD, all five independent loops (three on split 1, one each on splits 2 and 3) improved on the seed (abnormal EEG +0.054 to +0.091; neonatal seizure +0.048 to +0.087; MDD +0.057 to +0.141). ADHD on TDBRAIN has no repeats.

Pooled over 11 AUROC tasks (Table 3), the loop gained +0.050 over the seed ($p = 0.029$; $I^2 = 85\%$), +0.036 over best-of-20 ($p = 0.063$) and +0.017 over Optuna ($p = 0.28$). The gain over the seed is fragile: leaving out any one of six tasks (ADHD-200, MDD, ADHD on TDBRAIN, schizophrenia, abnormal EEG or neonatal seizure) gives $p = 0.054$–0.064; counting the two TDBRAIN tasks, which share controls, as one dataset gives +0.043 ($p = 0.062$); and leaving out TDBRAIN and ADHD-200 gives +0.027 ($p = 0.25$). The loop's advantage over Optuna stays at +0.017 to +0.021 and non-significant in every variant.

4.4 Shortcuts and demographic confounds

Under protocol v1, every ADHD-200, ABIDE and pooled-IBD champion referenced non-biological inputs: site, age and sex (ADHD-200); site, age, sex and head motion (ABIDE); study identity (IBD). These inputs carry substantial signal alone (Figure 2): site plus age and sex reach a pooled Eval AUROC of 0.775 on ADHD-200, close to the v1 champion (0.813), and study identity alone gives 0.693 on IBD. Rescored within site, the v1 ADHD-200 champion drops from 0.813 to 0.731, close to the v2 champion (0.717).

Table 3: Pooled Eval gain over the 11 AUROC tasks: equal-weight mean with t-interval (primary) and DerSimonian–Laird random effects (secondary). Sensitivity rows treat the TDBRAIN tasks (shared controls, separate recruitment, no transfer) as one dataset or drop them, and drop ADHD-200 (gain carried by one split). Leave-one-task-out results are in the released files.

Comparison	k	mean gain	95% CI	p	random effects [95% CI]	I^2
Loop vs. seed	11	+0.050	[+0.006, +0.094]	0.029	+0.044 [+0.009, +0.079]	85%
Optuna vs. seed	11	+0.033	[−0.004, +0.069]	0.073	+0.025 [−0.001, +0.052]	72%
Best-of-20 vs. seed	11	+0.014	[−0.003, +0.031]	0.095	+0.011 [−0.004, +0.025]	73%
One-shot vs. seed	11	+0.004	[−0.005, +0.013]	0.350	+0.002 [−0.003, +0.007]	16%
Loop vs. Optuna	11	+0.017	[−0.017, +0.051]	0.281	+0.015 [−0.004, +0.034]	67%
Loop vs. best-of-20	11	+0.036	[−0.002, +0.075]	0.063	+0.027 [+0.002, +0.053]	83%
Loop vs. one-shot	11	+0.046	[+0.003, +0.089]	0.037	+0.039 [+0.006, +0.071]	86%
Sensitivity of the equal-weight mean						
Loop vs. seed, TDBRAIN as one dataset	10	+0.043	[−0.003, +0.089]	0.062		
Loop vs. seed, without TDBRAIN	9	+0.034	[−0.013, +0.082]	0.130		
Loop vs. seed, without ADHD-200	10	+0.046	[−0.002, +0.094]	0.060		
Loop vs. seed, without TDBRAIN and ADHD-200	8	+0.027	[−0.024, +0.078]	0.250		
Loop vs. Optuna, TDBRAIN as one dataset	10	+0.019	[−0.020, +0.057]	0.299		
Loop vs. Optuna, without TDBRAIN	9	+0.020	[−0.023, +0.063]	0.319		
Loop vs. Optuna, without ADHD-200	10	+0.018	[−0.021, +0.056]	0.323		
Loop vs. Optuna, without TDBRAIN and ADHD-200	8	+0.021	[−0.030, +0.071]	0.365		

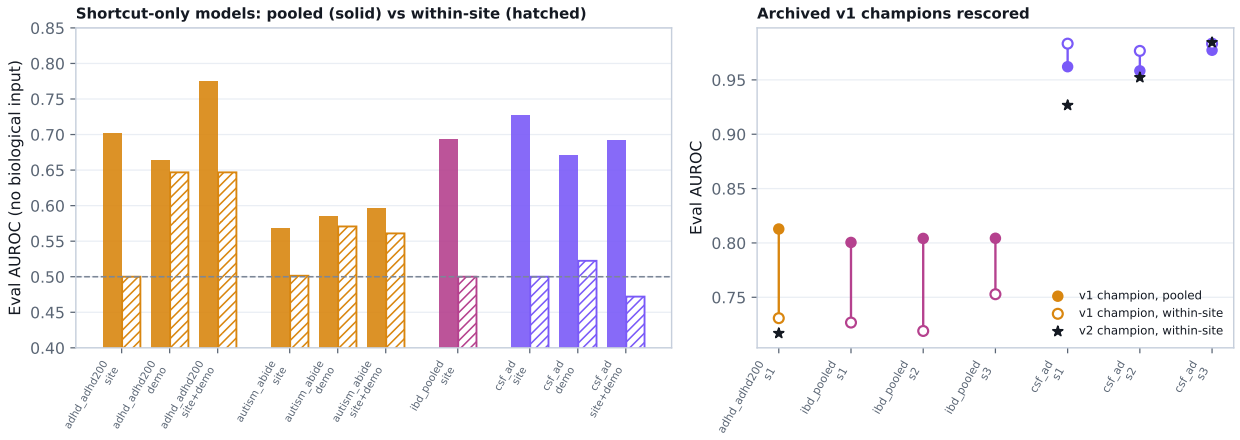


Figure 2: Left: Eval AUROC of shortcut-only models (no biological input), pooled (solid) and within site (hatched). Right: archived v1 champions rescored pooled and within site, with the v2 champion (no v2 point for IBD: the redefined v3 task is scored on different Eval samples).

Within-site scoring removes the site shortcut but not demographic signal, which brain data encode even when age and sex are not given as inputs. We therefore tested whether age and sex explain the v2 predictions (Table 6, Figure 5). On ADHD-200, age and sex alone reach a within-site AUROC of 0.645, close to the champion's 0.665; adjusting for them lowers the champion to 0.621 (seed 0.545), and the prediction adds significant information beyond age and sex in only one of three splits. On ABIDE and both TDBRAIN tasks, age and sex alone are at chance (0.45–0.51) and adjustment changes the AUROC by at most 0.021; the loop's MDD gain shrinks from +0.104 to +0.076 after adjustment and its ADHD (TDBRAIN) gain is unchanged (+0.140). Age and sex therefore do not explain the TDBRAIN results, but they cannot rule out other differences between separately recruited patients and controls (Section 4.5). ASEEG and our TUAB build carry no demographics.

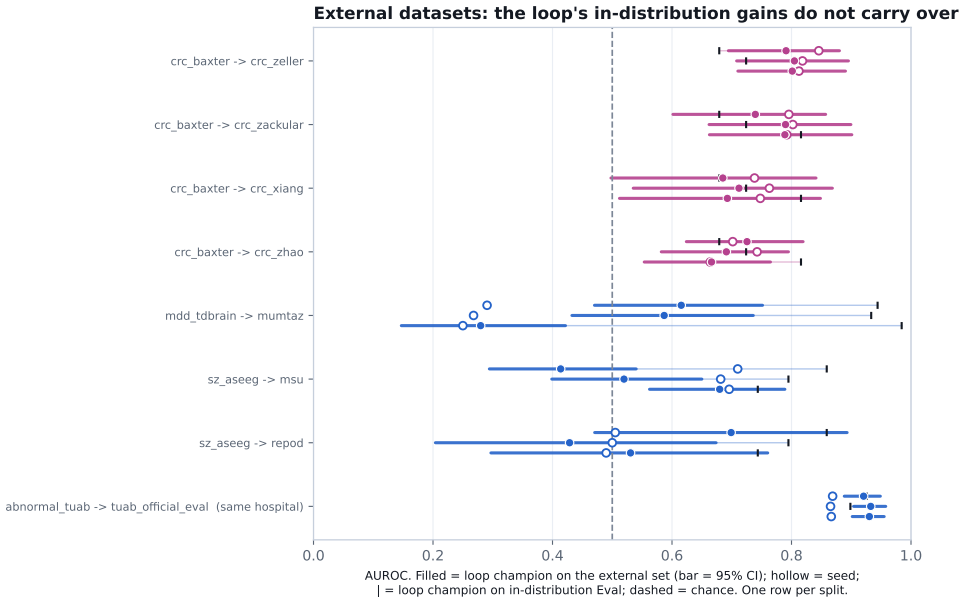


Figure 3: External datasets, absolute AUROC with 95% CI for the loop champion (filled) and the seed (hollow), one row per split; ticks mark the champion's in-distribution Eval score; dashed: chance.

4.5 In-distribution gains did not transfer to other cohorts

Applied unchanged to external data (Figure 3, Table 7), the TDBRAIN depression seed was *inverted* on the Mumtaz cohort (AUROC 0.25–0.29), and the champions were below or indistinguishable from chance in all three splits (0.28–0.62). The numerical gain moves an inverted model towards chance; it is not evidence of a transferable marker. The schizophrenia champion was worse than the seed on the MSU cohort in all three splits (0.696 → 0.538) and above chance in only one; on the small RepOD cohort neither seed nor champion was distinguishable from chance. The CRC champion stayed above chance on the four external cohorts in 11 of 12 split–cohort pairs, but beat the seed in only 2 of 12. Only abnormal EEG improved on its additional test set (AUROC 0.867 → 0.928, all CIs excluding chance), and that set is the official TUAB evaluation split from the same hospital and protocol: a within-hospital check, not transfer. Acquisition differences (amplifier, montage, recording length, age range) are an alternative explanation to biology for every failure to transfer.

4.6 Variance, transduction and leakage

Repeating the loop on the same split varied the Eval gain less than drawing a new split (run-to-run SD 0.005–0.044, median 0.017; split-to-split median 0.042; smaller for 8 of 9 tasks; Figure 7). The sign was stable across repeats on abnormal EEG, neonatal seizure, MDD and schizophrenia (all positive), negative on IBD in all three repeats, and flipped on CRC, CSF AD and drug response. All seeds were exactly inductive. Scoring Eval inside a reference batch built from the dev pool changed champion scores by −0.0006 on average (worst −0.024), so reported scores do not exploit Eval-set statistics (Figure 8); 11 champions output batch-relative ranks and would need a reference cohort to score single patients. Finally, every champion was rescored in network-isolated containers with subject identifiers replaced by order-preserving tokens; all scores were reproduced to within

4×10^{-4}, whereas a positive-control solution that read labels from identifiers dropped from AUROC 1.000 to 0.500. A static scan of all 2,010 saved solution files found no access to files, network or harness beyond the resources a task provides (Appendix A).

5 Discussion

When the loop helped. The clearest gains came on abnormal EEG and neonatal seizure detection: signal-processing tasks with large or well-annotated data and clinically meaningful labels. They were the only tasks where the loop beat a loop-informed Optuna search, the gains replicated in all five independent loops, and independent LLM proposals almost never beat the seed. The active ingredient is feedback, not the LLM's prior knowledge alone.

When it did not. Elsewhere the loop was not reliably better than ordinary search, and on four tasks it lost or stayed flat on held-out data in most splits despite consistent Dev gains. Its largest in-distribution gains, on TDBRAIN, were matched by Optuna, are not explained by age or sex, and did not transfer, which points to dataset-specific signal that any optimizer would find. Given site and demographics, the loop used them. A held-out set scored once, a non-LLM baseline, shortcut-only and confound models, and an external cohort each changed our reading of at least one task; the Dev curve alone would have changed none. Where a rough comparison with published results is possible, the champions were not better (Helsinki 0.899 vs. about 0.955 [35]; on CRC the seed already matched the MicrobiomeHD baseline [16]).

Limitations. Several Eval sets are small (20–43 subjects), the three splits overlap, and with three splits per-task inference across splits is weak; repeats cover 9 of 12 tasks. The v2 protocol, the IBD task definition and the Optuna search spaces were fixed after Eval results had been seen, and no confirmation on fresh splits with a pre-registered analysis was run. We used one LLM at one setting and a CPU budget that favours light models, and three early loops used an earlier brief wording. TDBRAIN cases and controls were recorded separately. Further caveats are in Appendices B and C.

6 Conclusion

An LLM autoresearch loop with feedback improved held-out performance over a simple seed on most of 12 biomedical tasks, clearly more than the same LLM without feedback, but significantly more than a loop-informed non-LLM search on only two signal-processing tasks, where its gains also replicated across independent runs. The same loop exploited site and demographic shortcuts when allowed, and its psychiatric-EEG gains did not transfer. Gains from such a loop are a hypothesis to test on held-out, shortcut-free and external data, not evidence of a better biomarker.

Data and code availability

Code, all loop records, champions, Eval predictions and robustness analyses: `https://github.com/radnyees/justautoeverything`. Raw data are not redistributed (TDBRAIN data use agreement, TUH/NEDC terms, CC BY-NC licences); `projects/*/data/` contains download and preprocessing scripts. Compute: $142.31 on Modal CPU containers (LLM calls not included).

A Additional methods

Prompt and rules. The brief asks for one well-motivated change per round, forbids reading anything outside what the harness passes in, requires inner CV (grouped by subject) for any tuning and lists the available libraries (numpy, scipy, scikit-learn, LightGBM, XGBoost, CatBoost, MNE, pyRiemann, nilearn, PyTorch on CPU, among others). Candidates run on Modal containers with 2–16 cores depending on the task.

Shortcut-only models. On the four multi-site tasks, L2 logistic regressions using only site or study one-hot codes and/or demographics were fit on the dev pool and scored on Eval with both the pooled (v1) and within-site (v2) metric.

Age/sex confounds. For tasks with per-subject age and sex (ADHD-200, ABIDE, TDBRAIN MDD and ADHD), we report the Eval AUROC of a model using age and sex only (fit on the dev pool); the AUROC of the seed or champion prediction after adjusting for age and sex (and site); whether the prediction tracks age or sex within controls; and a likelihood-ratio test of whether the prediction adds information about the label beyond age, sex and site.

Transduction audit. The Eval protocol featurizes dev-pool and Eval items in one call and passes the whole Eval batch to `fit_predict`. For every seed and champion we tested (i) whether dev-pool features change when Eval items are featurized with them; (ii) a *chunked* score, with Eval split into 20 chunks (10 for large tasks), each featurized with the dev pool and predicted on its own; and (iii) a *reference-batch* score, as (ii) but each chunk predicted inside a batch of dev pool plus chunk, so batch statistics come from training-time data.

Leakage audit. The loops' candidates were scored in containers that had network access and the data volume (including labels) mounted, and some datasets encode the label in subject identifiers. We therefore (i) rescored every champion, and the seeds of tasks whose identifiers encode the label, in network-isolated containers with real identifiers, with identifiers replaced by order-preserving tokens, and with random tokens; (ii) ran positive controls (a solution that reads the label from the identifier; a network probe); and (iii) statically scanned all 2,010 saved solution files for file, network, subprocess and harness access. Random tokens change scores by up to 0.034 on Eval because they change the assignment of subjects to inner-CV folds; order-preserving tokens reproduce every score. Rejected candidates of the 36 main loops were not saved and are covered only through their champions.

Pooled statistics. Drug response (Pearson r) is excluded from pooling. For random effects, task standard errors are floored at 0.005. Benjamini–Hochberg adjusted values [6] and leave-one-task-out estimates for every comparison are in the released files.

B Disclosures

Decisions made after Eval results had been seen. (i) The single-study IBD task (16 stool controls) was replaced by a pooled four-study task. (ii) Protocol v2 was introduced after v1 Eval results and v1 champion code had been seen; v2 loops were rerun on the same split seeds. (iii) Two Mumtaz subjects without eyes-closed data were excluded. (iv) The Optuna search spaces were written after all loop results. (v) The equal-weight pooled mean was made primary after

DerSimonian–Laird gave infinite weight to arms identical to the seed. (vi) A leakage in the pooled IBD task (biopsies and repeated samples of one person or co-twin could fall in both Dev and Eval) was found by our audit; the task was redefined as stool only, grouped by person or twin pair, and all IBD runs were redone. Every change that altered a task reran that task's loops from scratch.

Unequal conditions. The main loops of CRC, plasma AD and drug response started before protocol v2 and used the v1 wording of the brief (these tasks provide no demographics either way); their split-1 gains differ from the two v2-worded repeats by no more than run-to-run variation on other tasks. Claude Code CLI versions 2.1.287–2.1.296 were used; model and effort were identical. Two ABIDE loops stopped early (19 and 11 rounds).

Data-level choices. The 200 drugs were selected by variance across all cell lines, including later Eval lines. Proteomics intensities are as released, normalized across all samples by the original pipelines. Helsinki smoothing windows are centred (non-causal), acceptable offline but not for online detection. Subject identifiers that encode the label were visible to solutions; no solution read them (Appendix A). The seeds were written with an LLM coding assistant before any loop ran.

C Dataset caveats

- **TDBRAIN.** All controls are adults, so tasks are age-matched to the shared range (ADHD: adults only); age is never given to the model. Patients and controls were recruited and recorded separately, so state or protocol differences cannot be excluded.
- **ASEEG.** The channel order is undocumented and was inferred from data (its correlation structure matches labelled RepOD recordings: $r = 0.855$ for the standard 10–20 order, $r < 0.1$ for the alternatives). MSU is adolescents with 1-min, 16-channel recordings.
- **TUAB.** First 60 s skipped, next 5 min used, 19 channels at 100 Hz, clipped at $\pm 800\,\mu$V [18].
- **Helsinki.** Label = majority of three experts per second; published results often use leave-one-out or other annotation rules [20, 35].
- **ABIDE / ADHD-200.** Within-site AUROC is stricter than the pooled metrics usually reported [1, 19].
- **IBD.** Four MicrobiomeHD studies, stool samples only, grouped by person (by twin pair in the twin study); several people contribute more than one sample.
- **Proteomics.** Plasma: single centre, the only open labelled AD plasma MS set we found. CSF label = clinical diagnosis.
- **Drug response.** 200 most variable drugs ($\geq$70% coverage) from the PRISM secondary screen; per-drug Pearson r on held-out cell lines.

D Additional results

What the loop built. Figure 11 shows the feature and model families in the final champions as *declared by the LLM*; it is a self-report, not an analysis of the code. Champions typically extended the seed's representation with domain features: finer clinical sub-bands, aperiodic slope,

Table 4: Seed vs. loop champion per task (means over 3 splits), Dev and Eval. "improved": splits with Eval gain > 0; "sig.": splits with one-sided per-split bootstrap $p < 0.05$. Symbols as in Table 2.

		Dev (5-fold CV)		Held-out Eval			
Modality	Task	seed	champion	seed	champion	Eval gain (mean ± sd)	improved / sig.
EEG	Abnormal EEG (TUAB)	0.865	0.934	0.847	0.918	+0.071 ± 0.018	3/3 / 3/3
	MDD (TDBRAIN)	0.827	0.927	0.850	0.954	+0.104 ± 0.043	3/3 / 3/3
	ADHD (TDBRAIN)	0.731	0.882	0.785	0.922	+0.137 ± 0.095	3/3 / 2/3
	Schizophrenia (ASEEG)	0.806	0.883	0.677	0.799	+0.122 ± 0.084	3/3 / 1/3
	Neonatal seizure (Helsinki)	0.744	0.858	0.827	0.899	+0.072 ± 0.021	3/3 / 3/3
fMRI	Autism (ABIDE I)†	0.740	0.787	0.781	0.778	−0.002 ± 0.028	1/3 / 0/3
	ADHD (ADHD-200)†§	0.591	0.666	0.572	0.665	+0.093 ± 0.148	3/3 / 1/3
16S	CRC (Baxter)	0.755	0.789	0.795	0.740	−0.055 ± 0.035	0/3 / 0/3
	IBD (4 studies)†	0.725	0.802	0.721	0.735	+0.014 ± 0.071	1/3 / 1/3
Proteomics	AD, plasma	0.884	0.939	0.912	0.866	−0.046 ± 0.042	0/3 / 0/3
	AD, CSF†	0.859	0.943	0.913	0.954	+0.041 ± 0.068	2/3 / 1/3
Expression	Drug response (PRISM)‡	0.138	0.156	0.149	0.160	+0.012 ± 0.002	3/3 / 0/3

Table 5: Per-task Eval differences (mean over 3 splits) with Holm-adjusted two-sided p-values across 12 tasks from the hierarchical (hier.) and subject (subj.) bootstrap. Bold gain: hierarchical $p_{\text{Holm}} < 0.05$; ^: subject-bootstrap $p_{\text{Holm}} < 0.05$ only.

	Loop − seed			Optuna − seed			Loop − best-of-20			Loop − Optuna		
Task	gain	hier.	subj.	gain	hier.	subj.	gain	hier.	subj.	gain	hier.	subj.
Abnormal EEG (TUAB)	**+0.071**	0.012	0.012	**+0.042**	0.012	0.012	+0.031^	0.576	0.012	**+0.029**	0.012	0.012
MDD (TDBRAIN)	**+0.104**	0.012	0.012	**+0.096**	0.012	0.012	+0.070^	0.090	0.048	+0.008	1.000	1.000
ADHD (TDBRAIN)	**+0.137**	0.036	0.012	+0.132	0.060	0.090	**+0.137**	0.044	0.012	+0.004	1.000	1.000
Schizophrenia (ASEEG)	+0.122	0.480	0.165	+0.001	1.000	1.000	+0.094	0.576	0.182	+0.121	0.440	0.504
Neonatal seizure (Helsinki)	**+0.072**	0.012	0.012	−0.003	1.000	1.000	**+0.072**	0.012	0.012	**+0.075**	0.012	0.012
Autism (ABIDE I)†	−0.002	1.000	1.000	+0.001	1.000	1.000	−0.011	1.000	0.572	−0.003	1.000	1.000
ADHD (ADHD-200)†§	+0.093^	0.935	0.040	+0.078	1.000	0.351	+0.038	0.816	0.222	+0.015	1.000	1.000
CRC (Baxter)	−0.055^	0.480	0.040	−0.036	1.000	0.568	−0.048^	0.459	0.045	−0.020	1.000	1.000
IBD (4 studies)†	+0.014	1.000	1.000	−0.034	1.000	1.000	−0.015	1.000	0.572	+0.047	1.000	0.630
AD, plasma	−0.046	0.930	0.312	+0.023	1.000	1.000	−0.031	1.000	0.344	−0.068	1.000	0.220
AD, CSF†	+0.041	0.935	0.777	+0.058	1.000	1.000	+0.063	0.816	0.344	−0.017	1.000	1.000
Drug response (PRISM)‡	+0.012	0.935	0.162	−0.001	1.000	1.000	+0.012	0.944	0.260	+0.013	1.000	0.306

Table 6: Age and sex (mean over 3 splits; within-site AUROC for ADHD-200 and ABIDE). "age+sex only": model using demographics only; "adjusted": prediction adjusted for age, sex (and site); LR test: splits where the prediction adds information about the label beyond age, sex and site.

Task	arm	AUROC	age+sex only	adjusted for age+sex	LR test $p < 0.05$ (splits)
ADHD (ADHD-200)†	seed	0.572	0.645	0.545	0/3
	champion	0.665	0.645	0.621	1/3
Autism (ABIDE I)†	seed	0.781	0.487	0.777	3/3
	champion	0.778	0.487	0.774	3/3
MDD (TDBRAIN)	seed	0.850	0.454	0.871	3/3
	champion	0.954	0.454	0.947	3/3
ADHD (TDBRAIN)	seed	0.785	0.505	0.769	2/3
	champion	0.922	0.505	0.909	3/3

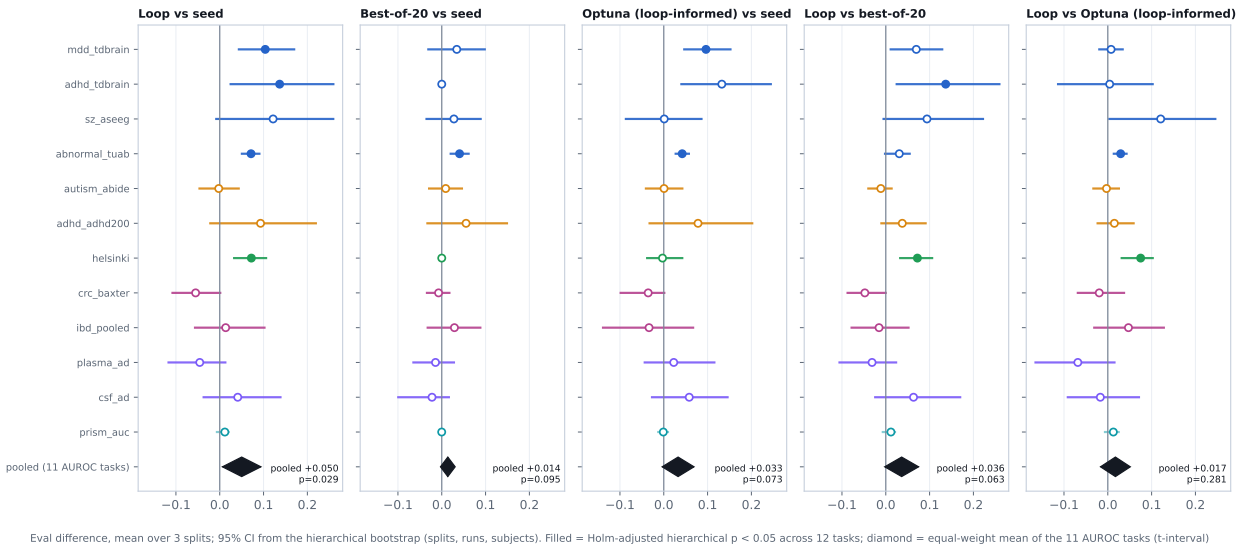


Figure 4: Per-task Eval differences with 95% CIs for five comparisons. Diamond: equal-weight mean of the 11 AUROC tasks.

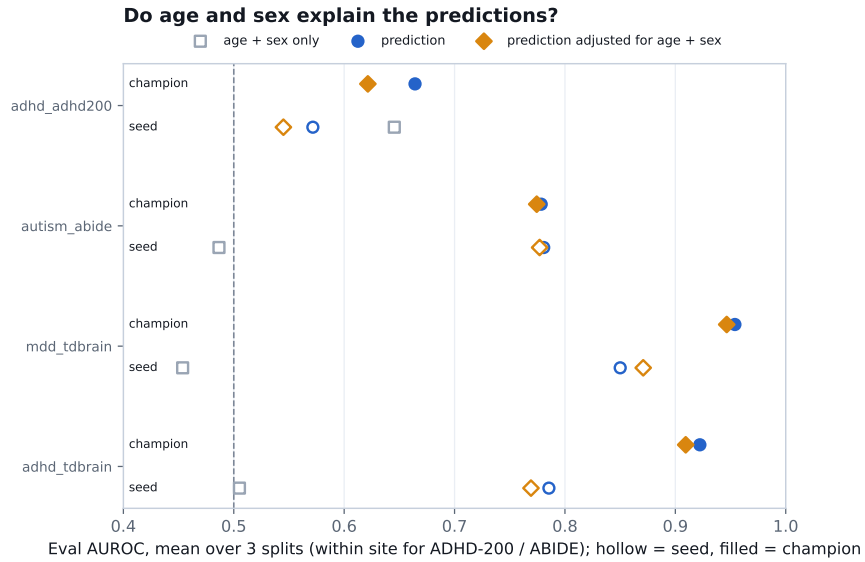


Figure 5: Eval AUROC of the prediction (blue), the prediction adjusted for age and sex (orange) and age and sex alone (grey). Hollow: seed; filled: champion; dashed: chance.

connectivity, complexity and Riemannian covariance features (EEG); per-neonate and per-channel background normalization and multi-scale temporal context (neonatal seizures); log-ratio transforms and batch correction (microbiome); known AD proteins and pathway aggregates (proteomics). Ensembles of a linear model with boosting or kernel methods were common.

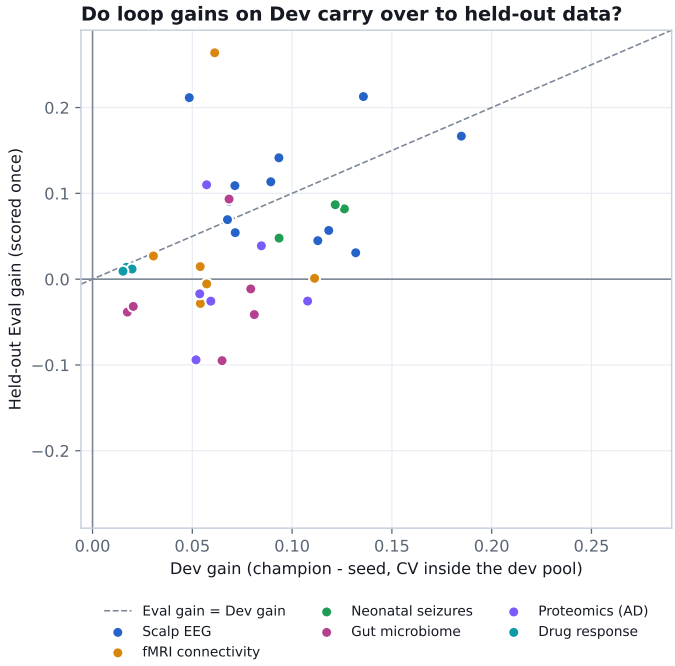


Figure 6: Dev gain vs. held-out Eval gain, one point per task and split. Dashed: Eval gain = Dev gain.

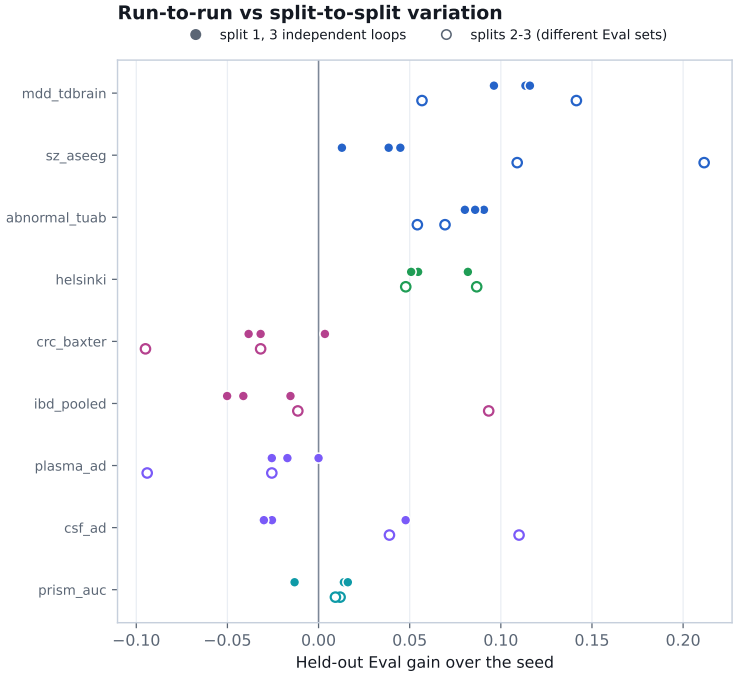


Figure 7: Eval gain of three independent loops on split 1 (filled) and of the loops on splits 2–3 (hollow), for the nine tasks with repeats.

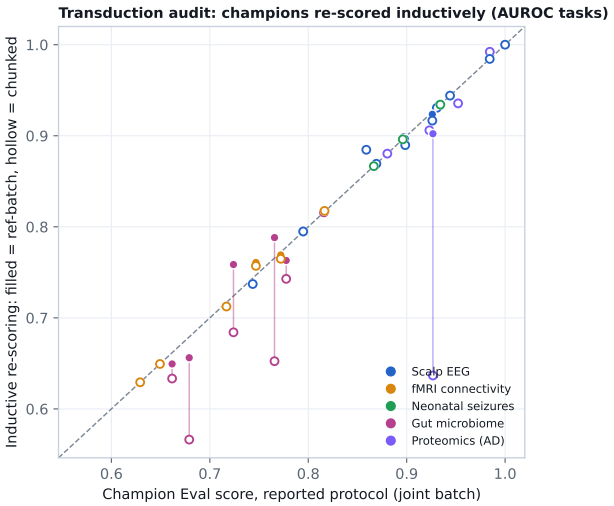


Figure 8: Transduction audit of the champions (AUROC tasks). x: reported Eval score (joint batch). Filled: reference-batch score; hollow: chunks scored in isolation.

Table 7: External datasets: seed and champion applied unchanged; 95% CI from the paired bootstrap.

Task	External set	split	seed	champion	gain	95% CI
Abnormal EEG (TUAB)	TUAB official eval	1	0.867	0.930	+0.064	[+0.034, +0.096]
		2	0.865	0.933	+0.067	[+0.035, +0.104]
		3	0.869	0.921	+0.052	[+0.020, +0.084]
MDD (TDBRAIN)	Mumtaz 2016	1	0.250	0.280	+0.030	[−0.123, +0.190]
		2	0.268	0.587	+0.319	[+0.120, +0.518]
		3	0.290	0.615	+0.325	[+0.154, +0.502]
Schizophrenia (ASEEG)	MSU	1	0.696	0.680	−0.016	[−0.108, +0.077]
		2	0.681	0.520	−0.162	[−0.276, −0.043]
		3	0.710	0.414	−0.296	[−0.438, −0.147]
	RepOD	1	0.490	0.531	+0.041	[−0.182, +0.262]
		2	0.500	0.429	−0.071	[−0.231, +0.109]
		3	0.505	0.699	+0.194	[−0.052, +0.448]
CRC (Baxter)	Zeller	1	0.813	0.801	−0.011	[−0.058, +0.035]
		2	0.818	0.805	−0.013	[−0.054, +0.024]
		3	0.846	0.791	−0.055	[−0.114, −0.002]
	Zackular	1	0.792	0.789	−0.003	[−0.053, +0.053]
		2	0.802	0.790	−0.012	[−0.055, +0.034]
		3	0.796	0.739	−0.056	[−0.144, +0.024]
	Xiang	1	0.748	0.693	−0.055	[−0.126, −0.007]
		2	0.763	0.712	−0.051	[−0.163, +0.054]
		3	0.738	0.685	−0.053	[−0.133, +0.012]
	Zhao	1	0.664	0.666	+0.002	[−0.029, +0.031]
		2	0.742	0.691	−0.051	[−0.112, +0.006]
		3	0.702	0.726	+0.024	[−0.043, +0.091]

Table 8: Every run. "failed" = timeouts + crashes. Eval gain CI and one-sided p from the per-split paired bootstrap (2,000 resamples).

Task	split	rounds	kept	failed	Dev gain	Eval gain	95% CI	p (one-sided)
Abnormal EEG (TUAB)	1	20	11	0	+0.068	+0.091	[+0.067, +0.116]	<0.001
	2	20	9	1	+0.068	+0.069	[+0.045, +0.096]	<0.001
	3	20	10	3	+0.071	+0.054	[+0.033, +0.076]	<0.001
MDD (TDBRAIN)	1	20	7	0	+0.089	+0.113	[+0.033, +0.203]	<0.001
	2	20	11	0	+0.093	+0.141	[+0.038, +0.258]	0.001
	3	20	6	0	+0.118	+0.057	[−0.005, +0.132]	0.042
ADHD (TDBRAIN)	1	20	6	0	+0.132	+0.031	[−0.031, +0.118]	0.186
	2	20	10	0	+0.185	+0.167	[+0.060, +0.303]	<0.001
	3	20	11	0	+0.136	+0.213	[+0.045, +0.392]	0.004
Schizophrenia (ASEEG)	1	20	6	0	+0.113	+0.045	[−0.104, +0.201]	0.307
	2	20	10	0	+0.048	+0.212	[+0.035, +0.420]	0.012
	3	20	8	0	+0.071	+0.109	[−0.019, +0.292]	0.065
Neonatal seizure (Helsinki)	1	20	11	0	+0.126	+0.082	[+0.034, +0.121]	0.002
	2	20	10	0	+0.093	+0.048	[+0.015, +0.106]	0.003
	3	20	7	1	+0.122	+0.087	[+0.045, +0.171]	<0.001
Autism (ABIDE I)[†]	1	19	7	1	+0.054	−0.028	[−0.091, +0.033]	0.816
	2	11	1	0	+0.031	+0.027	[−0.038, +0.094]	0.210
	3	20	10	0	+0.057	−0.005	[−0.067, +0.055]	0.564
ADHD (ADHD-200)[†§]	1	20	5	0	+0.061	+0.264	[+0.129, +0.344]	<0.001
	2	20	5	0	+0.111	+0.001	[−0.075, +0.081]	0.491
	3	20	7	0	+0.054	+0.015	[−0.062, +0.113]	0.306
CRC (Baxter)	1	20	5	0	+0.017	−0.038	[−0.096, +0.019]	0.922
	2	20	6	0	+0.020	−0.032	[−0.091, +0.021]	0.873
	3	20	10	0	+0.065	−0.095	[−0.185, −0.012]	0.991
IBD (4 studies)[†]	1	20	6	0	+0.081	−0.041	[−0.109, +0.024]	0.905
	2	20	4	0	+0.068	+0.093	[−0.006, +0.189]	0.031
	3	20	10	0	+0.079	−0.011	[−0.069, +0.053]	0.659
AD, plasma	1	20	6	0	+0.054	−0.017	[−0.106, +0.038]	0.790
	2	20	8	0	+0.052	−0.094	[−0.217, +0.000]	0.999
	3	20	6	0	+0.059	−0.026	[−0.125, +0.067]	0.772
AD, CSF[†]	1	20	10	0	+0.108	−0.026	[−0.155, +0.096]	0.692
	2	20	6	1	+0.085	+0.039	[−0.068, +0.168]	0.257
	3	20	8	0	+0.057	+0.110	[+0.014, +0.250]	0.015
Drug response (PRISM)[‡]	1	20	9	0	+0.017	+0.014	[−0.004, +0.032]	0.055
	2	20	9	0	+0.020	+0.012	[−0.003, +0.027]	0.059
	3	20	10	0	+0.015	+0.009	[−0.008, +0.026]	0.174

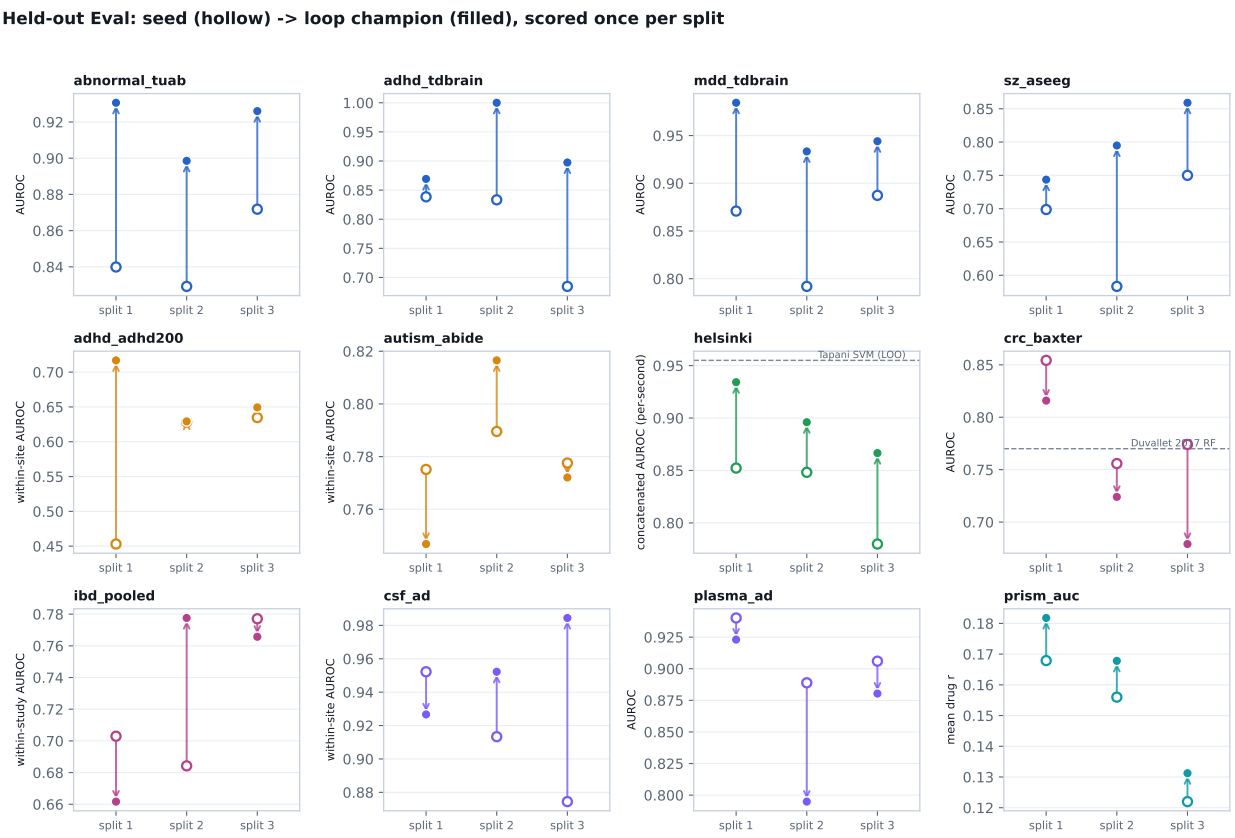


Figure 9: Held-out Eval score of the seed (hollow) and the loop champion (filled), per split. Dashed: published results on the same dataset (different protocols).

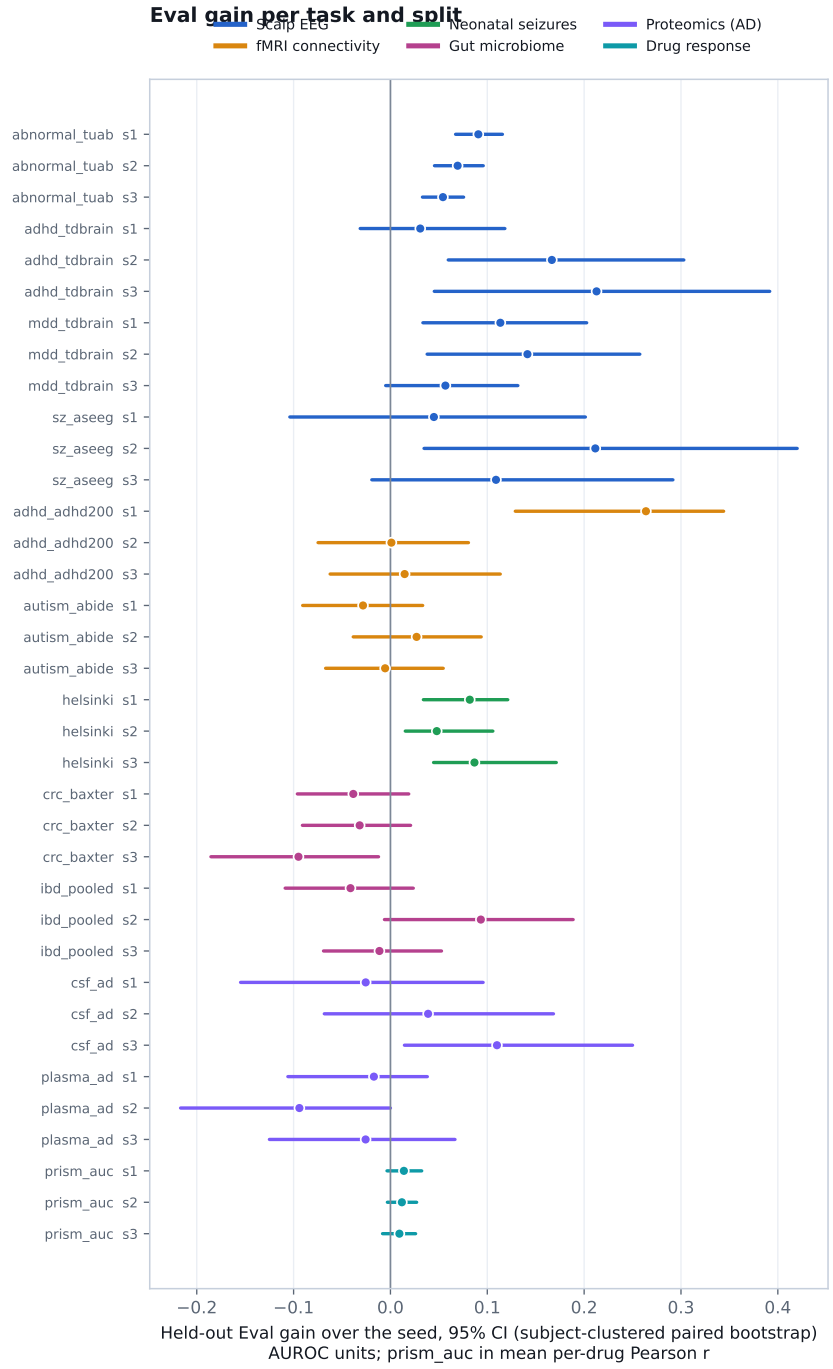


Figure 10: Held-out Eval gain over the seed with 95% CI (per-split paired bootstrap), per task and split.

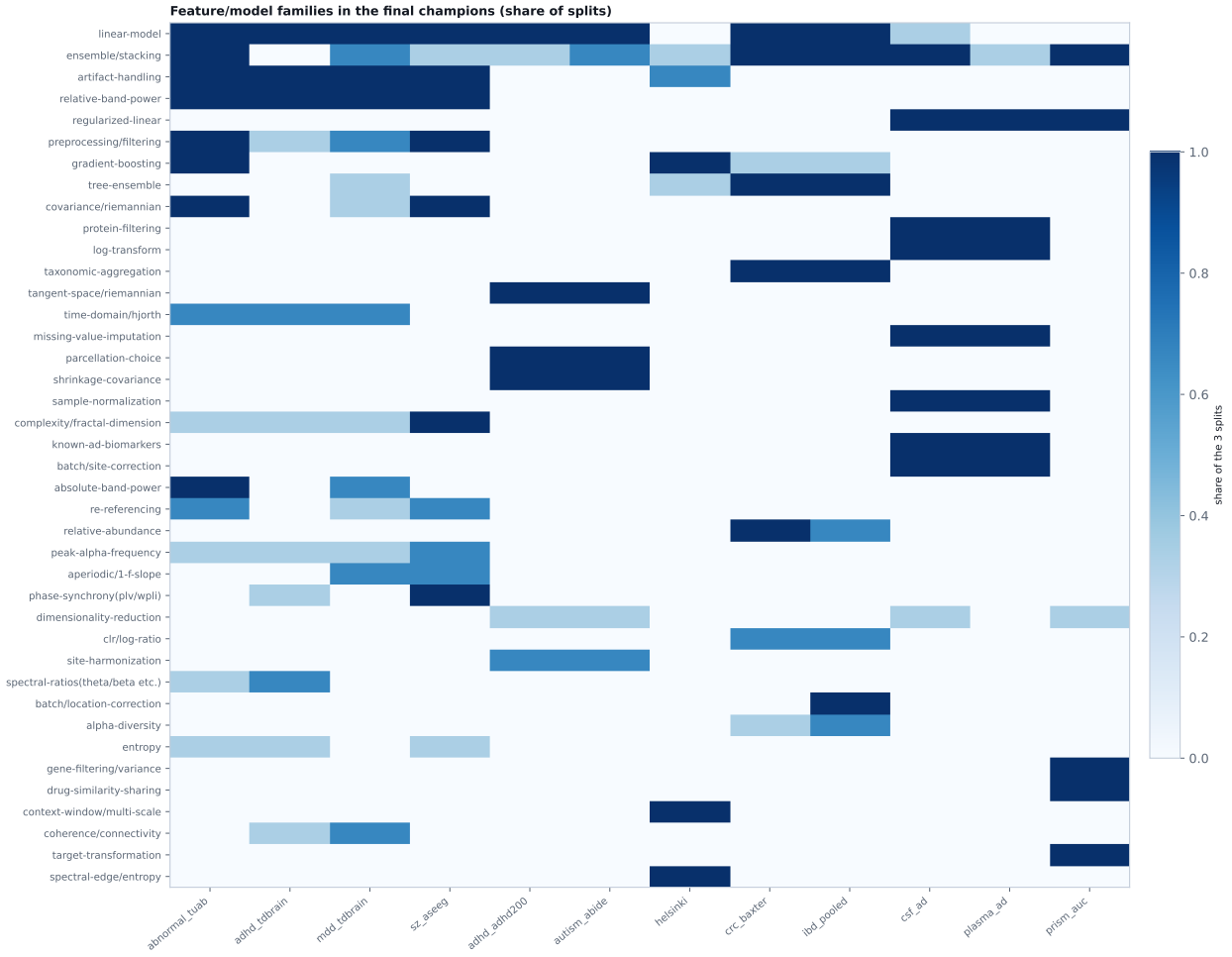


Figure 11: Feature/model families in the final champions (share of the 3 splits), as declared by the LLM.

E Loop trajectories

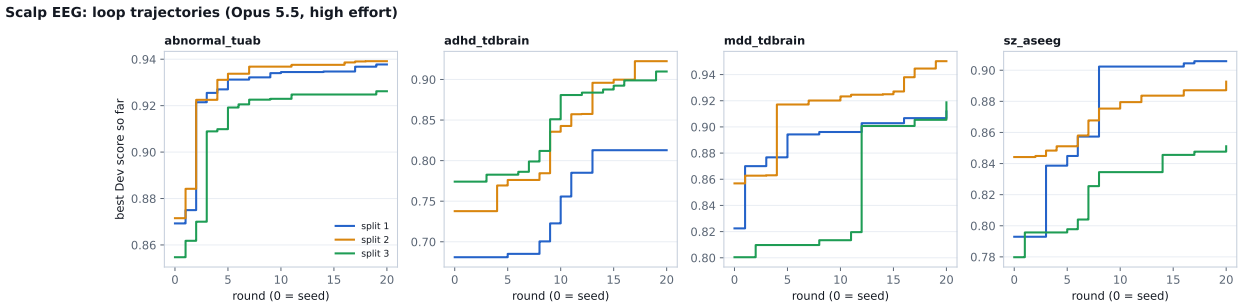


Figure 12: Best Dev score so far per round, scalp EEG tasks.

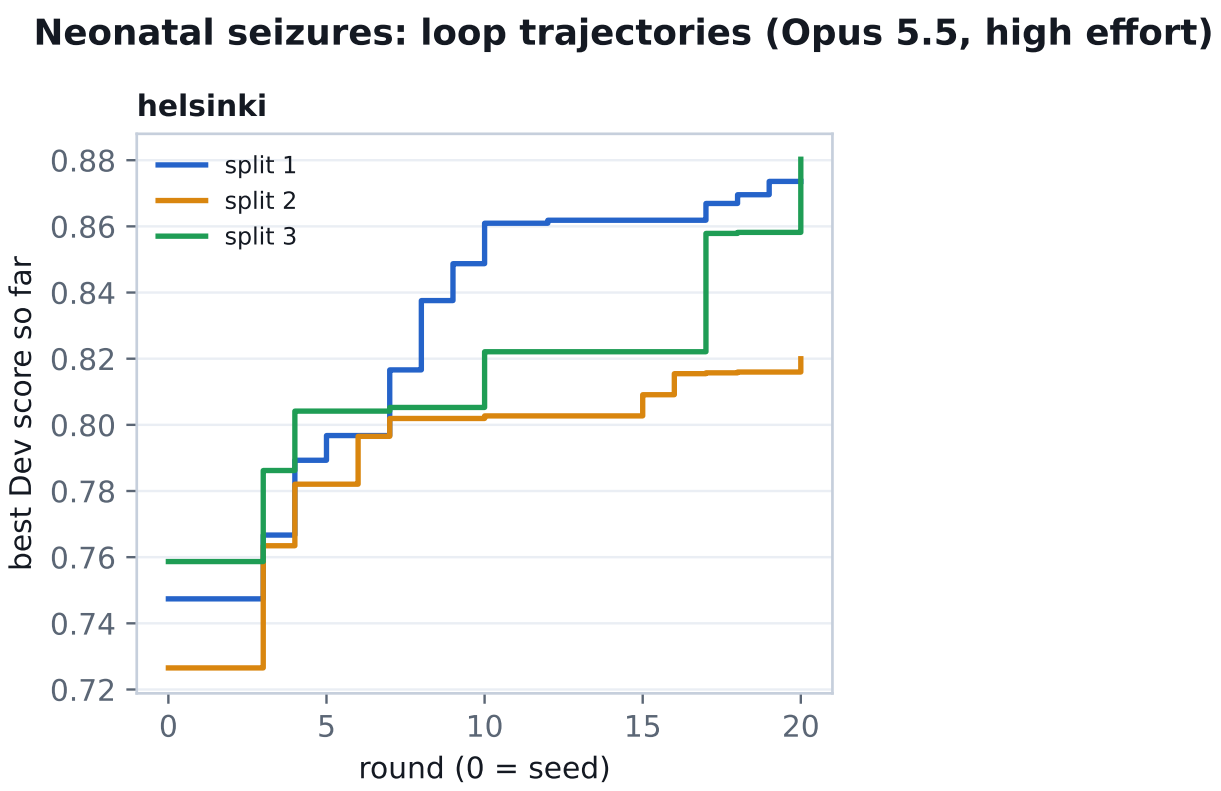


Figure 13: Best Dev score so far per round, neonatal seizure detection.

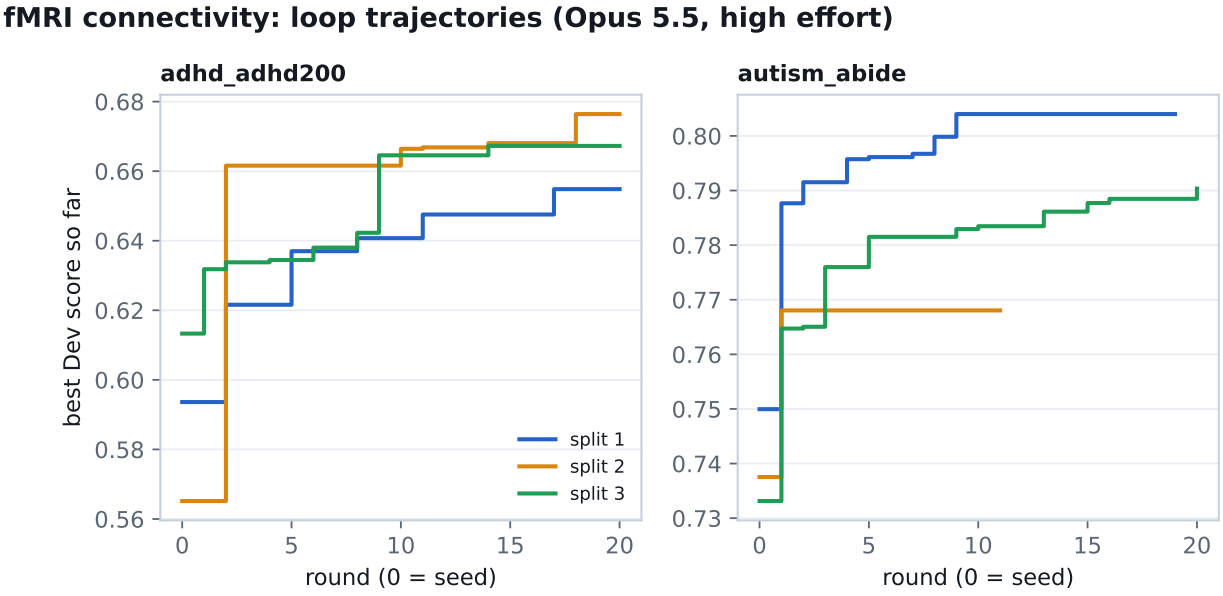


Figure 14: Best Dev score so far per round, fMRI connectivity tasks.

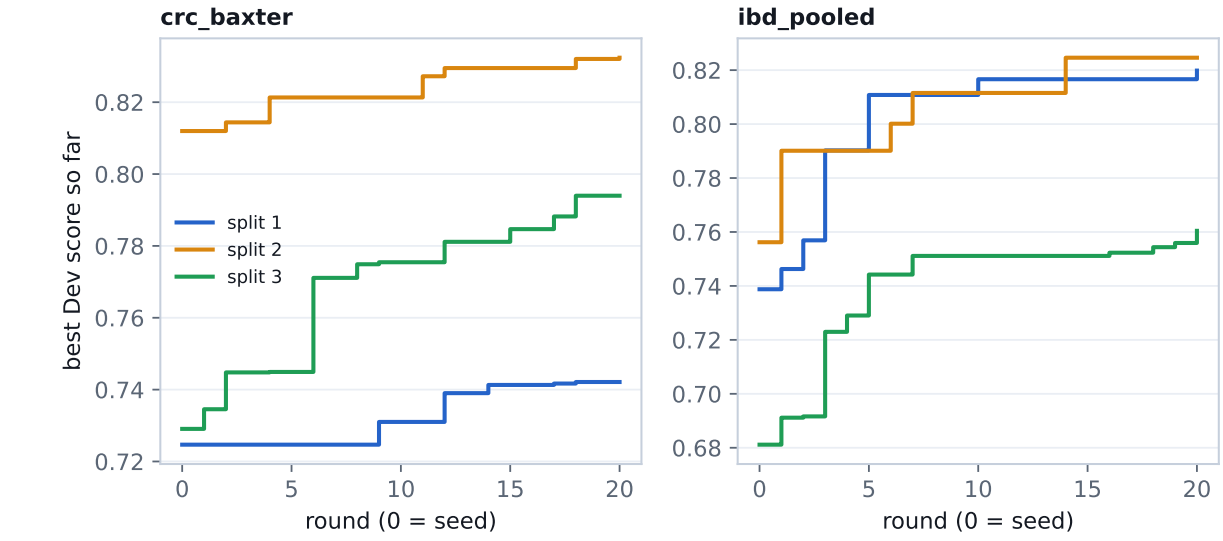


Figure 15: Best Dev score so far per round, gut microbiome tasks.

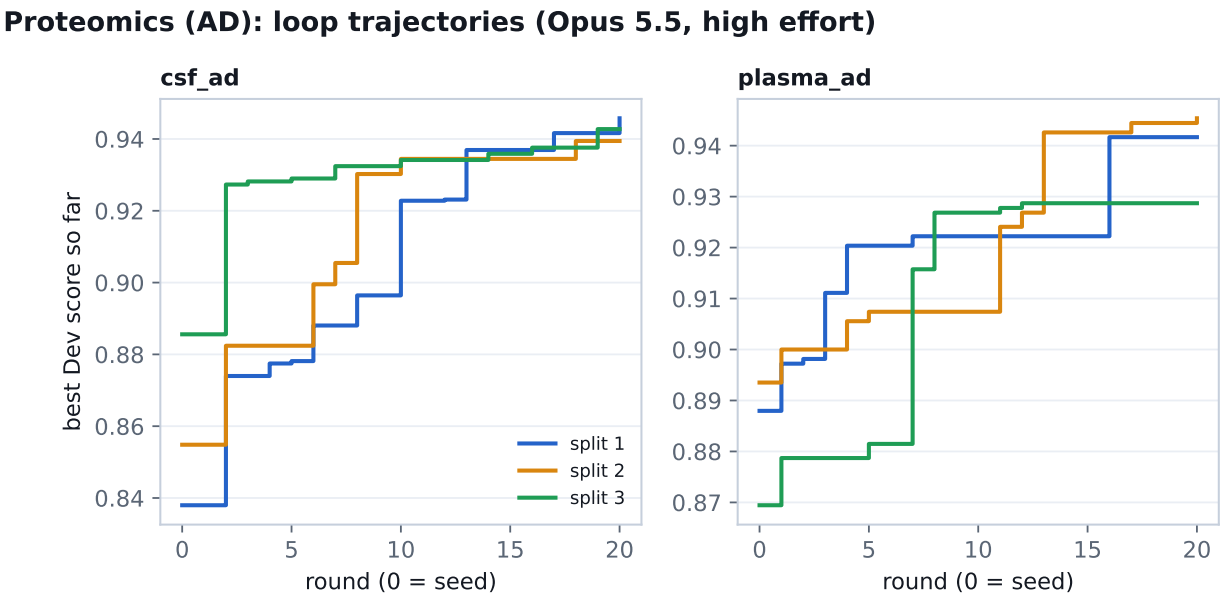


Figure 16: Best Dev score so far per round, proteomics (AD) tasks.

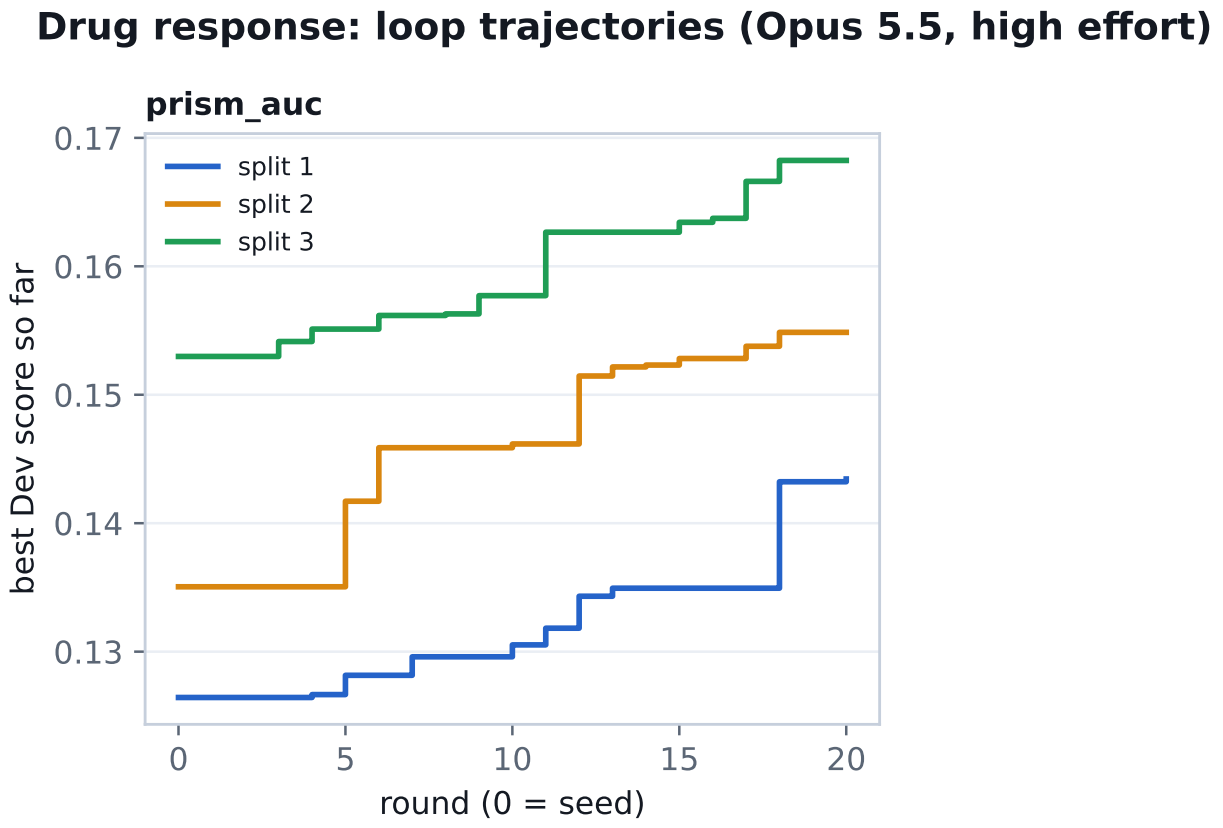


Figure 17: Best Dev score so far per round, drug response.

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